

Error Detection and Correction

Q1. Find the Hamming distance between:

- a) 000000 and 110011
- b) 1010101 and 1111000

Q2. For the codewords {000, 011, 101, 110}:

- 1. Find $d_{\min}$
- 2. Determine detection capability
- 3. Can it correct errors?

Q3. A block code has $d_{\min} = 5$.

- 1. How many errors can it detect?
- 2. How many errors can it correct?

Q4. Generate even parity for data 1011011. Check if 10110110 has error.

Q5. Generate 2D parity (even) for:

1 0 1 1
0 1 0 1
1 1 1 0

Q6. In 2D parity, row 3 and column 2 fail. Locate error. Can it correct double errors?

Q7. Encode dataword 1011 using Hamming $C(7,4)$.

Q8. Received codeword 0111011. Find syndrome, error position, correct it, extract data.

Q9. For Hamming code $n = 2^m - 1$. If minimum data bits required = 10, find m , n , k .

Q10. Data = 1101011, Generator = 1011. Perform CRC and find transmitted frame.

Q11. For $g(x) = x^3 + x + 1$:

1. Binary representation
2. Does it detect all single-bit errors?

Q12. If generator contains $(x+1)$, what errors are always detected?

Q13. If $r = 4$ in CRC:

1. Burst errors guaranteed detected?
2. Detection probability for $L = r+1$?

Q14. Find 1's complement checksum for:

1010

0111

1100

Q15. Represent -9 in 1's complement using 5 bits.

Solution Manual: Error Detection and Correction

Q1 Solution:

- a) Distance = 4
- b) Distance = 3

Q2 Solution:

All pair distances = 2

So $d_{\min} = 2$

Detection capability = 1 error

Correction = Not possible

Q3 Solution:

Detection = $d_{\min} - 1 = 4$ errors

Correction = $\text{floor}((d_{\min} - 1)/2) = 2$ errors

Q4 Solution:

Number of 1s = 5 parity bit = 1

Transmitted = 10110111

Received 10110110 error detected

Q5 Solution:

Row parity bits: 1,0,1

Column parity bits: 0,0,0,0

Construct final 4x5 block accordingly.

Q6 Solution:

Error at intersection (Row 3, Column 2)

Cannot correct double-bit errors reliably.

Q7 Solution:

Encoded codeword = 1011010 (using standard C(7,4))

Q8 Solution:

Syndrome $\neq$ 000 error position found via binary value

Corrected dataword extracted.

Q9 Solution:

Try $m=4$ $n=15$, $k=11$ satisfies $k \geq 10$

So $m=4$, $n=15$, $k=11$

Q10 Solution:

Append 3 zeros divide remainder obtained

Transmitted frame = data + remainder

Q11 Solution:

Binary = 1011

Yes, detects all single-bit errors since more than one term and $x_0=1$

Q12 Solution:

Detects all odd-number of bit errors

Q13 Solution:

All burst errors $L \leq 4$ detected

For $L=5$ probability = $1 - (1/2)^{(r-1)} = 1 - 1/8$

Q14 Solution:

Binary sum wrap carry complement checksum obtained

Q15 Solution:

+9 = 01001 complement = 10110